

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Supplementary Winter Examination – 2024					
Course: B. Tech		Branch: Petrochemical Engineering		Semester: III	
Subject Code & Name: BTPCC304 Petrochemical Engineering -I					
Max Marks: 60		Date: 12/02/2025		Duration: 3 Hr.	
Instructions to the Students:					
1) Each question carries 12 marks.					
2) Question No. 1 will be compulsory and include objective-type questions.					
3) Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.					
4) The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.					
5) Use of non-programmable scientific calculators is allowed.					
6) Assume suitable data wherever necessary and mention it clearly.					
				(Level/CO)	Mark s
Q.1	Objective type questions. (Compulsory Question)				12
1	_____ is the form of Sulfur compound present in crude oil.			R/CO1	1
	a) Mercaptans	b) Thiol	c) Monosulfide		
2	Natural gas mostly contains _____			U/CO1	1
	a) Ethane	b) Propane	c) Methane		
3	_____ is the type of floaters drilling rig.			R/CO1	1
	a) Drill Ship	b) Semi-Submersible	c) Barges		
4	Light crude oil have specific gravities _____			U/CO1	1
	a) <0.825	b) >0.825	c) 0.825-1		
5	By Visbreaking process, Viscosity becomes _____			A/CO1	1
	a) decreases	b) increases	c) remain same		
6	Which of the following is Not a product of atmospheric distillation unit?			A/CO1	1
	a) Gasoline	b) Kerosene	c) Lube oil		
7	Primary distillation is _____			R/CO1	1
	a) Vacuum distillation	b) Pressure distillation	c) Atmospheric distillation		
8	Octane number is an important test for _____			A/CO4	1
	a) Gasoline	b) Kerosene	c) Lube oil		
9	_____ has high antiknock value.			U/CO5	1
	a) Methanol	b) Ethanol	c) Isopropanol		
10	Diesel Engine is based on _____			A/CO5	1
	a) Spark ignition	b) Compression ignition	c) Both		
11	Asphalt is produced from _____.			U/CO5	1
	a) Vacuum distillation	b) Pressure distillation	c) Atmospheric distillation		
12	Deasphalting is usually utilized after _____ unit .			R/CO5	1
	a)Atmospheric distillation	b) Pressure distillation	c) Vacuum distillation		

Q 2	Solve the following.		12
A)	Write a note on following compounds present in crude oil. 1) Paraffin 2) Naphthenes 3) Aromatics	R, U/ CO1	6
B)	Describe the ultimate elemental analysis of petroleum. How Porosity and Permeability properties play important role in reservoir rocks.	R, U/ CO1	6
Q. 3	Solve the following.		12
A)	Explain the types of Drilling rig with its diagram.	U, A/ CO3	6
B)	Give the note on : Selection of Hydrocarbon in a Gasoline Blends	R, U /CO5	6
Q. 4	Solve Any Two of the following.		12
A)	Give the Continuous stage wise distillation diagram. Explain the Vapor/Liquid Equilibrium with the help of suitable equation.	A/CO4	6
B)	What is the need of dehydration and desalting of crude oil? Explain in detail electric desalting of crude oil with diagram.	R/CO4	6
C)	List out any four conversion processes used in the refinery. Explain Visbreaking and refinery flow diagram with visbreaking unit.	R, U/ CO3	6
Q. 5	Solve Any Two of the following.		12
A)	Give the detailed note on following: 1) Specific gravity of crude oil 2) Cloud and pour point of crude oil	R,U/ CO5	6
B)	Explain the Primary classification of reservoir rocks.	R, U/ CO1	6
C)	Describe various Drilling methods in detail.	U, A/ CO2	6
Q. 6	Solve Any Two of the following.		12
A)	What is ment by Drilling Mud? List out functions of drilling mud.	R, U/ CO2	6
B)	Enlist Functions of Geologist on drilling well.	U, A/ CO3	6
C)	Explain the Origin of Petroleum. Draw structural diagram of oil traps.	U,A/ CO3	6
	*** End ***		